

*** Multiply 110(6) with 101(5) ***

Weight	2^5	2^4	2^3	2^2	2^1	2^0
				1	1	0
x			1	1	0	1
				1	1	0
		0	0	0	0	X

$$\begin{array}{r} 6 \\ \times 5 \\ \hline 30 \end{array}$$

Weight	2^5	2^4	2^3	2^2	2^1	2^0
				1	1	0
x			1	1	0	1
				1	1	0
		0	0	0	0	X

$$\begin{array}{r} 6 \\ \times 5 \\ \hline 30 \end{array}$$

Fractional Multiplication

In the binary, two numbers are 110.01 and 11.10. In the first number (110.01), the number of digits after the binary point are 2. Similarly, in the second number (11.10), the number of digits after the binary point are 1.

$$\begin{array}{r} 110.01 \\ \times 11.10 \\ \hline \end{array} \quad \begin{array}{l} \rightarrow 2 \\ \rightarrow 1 \end{array} \quad \begin{array}{r} 6.25 \\ \times 3.5 \\ \hline 21.875 \end{array}$$

$$\begin{array}{r} 110.01 \\ \times 11.10 \\ \hline \end{array} \quad \rightarrow \quad \begin{array}{r} 11001 \\ \times 111 \\ \hline \end{array}$$

After multiplying the two numbers, the result will be **10101111**.

$$\begin{array}{r} 11001 \\ \times 111 \\ \hline 111001 \\ + 111001\text{X} \\ + 11001\text{X X} \\ \hline 10101111 \end{array}$$

Weight	2^5	2^4	2^3	2^2	2^1	2^0
				1	1	0
x				1	0	1
				1	1	0
		0	0	0	0	X
	1	1	0	X	X	

$$\begin{array}{r}
 6 \\
 \times 5 \\
 \hline
 30
 \end{array}$$

Weight	2^5	2^4	2^3	2^2	2^1	2^0
				1	1	0
x				1	0	1
				1	1	0
		0	0	0	0	X
	1	1	0	X	X	
	1	1	1	1	0	

$$\begin{array}{r}
 6 \\
 \times 5 \\
 \hline
 30
 \end{array}$$

Now, we just need to put the binary point at the appropriate location. In the first number 6.25 (110.01) the number of digits after the binary point are 2. And similarly, in the second number 3.5 (11.1), the number of digits after the binary point are 1. The total number of digits are 3 (2 + 1). Therefore, in the obtained result, put the binary point after the 3 digits from the LSB.

[illegible]

And therefore, the result after the multiplication is **10101.111**. If we see its binary equivalent then it is equal to **21.875**. In this way, we can multiply two fractional binary numbers.

2-BIT MULTIPLIER

$$A = \begin{matrix} 2^2 & 2^1 & 2^0 \\ A_2 & A_1 & A_0 \end{matrix}$$

$$B = \begin{matrix} B_1 & B_0 \end{matrix}$$

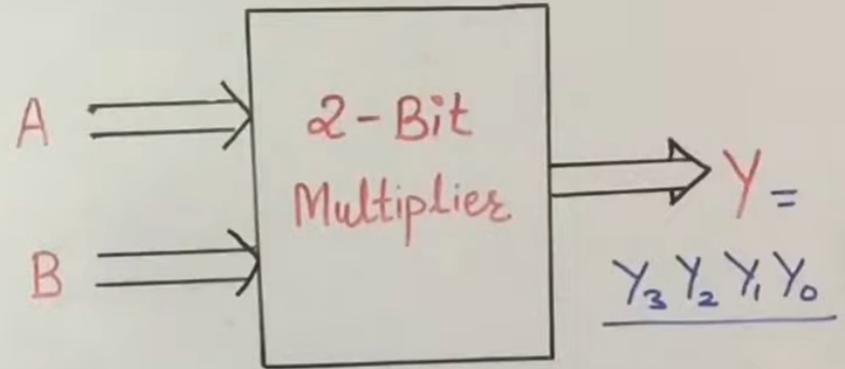
	$A_1 B_0$	$A_0 B_0$
$+ \begin{matrix} C_1 \\ \downarrow \end{matrix}$	$A_1 B_1$	$A_0 B_1$
$+ \begin{matrix} C_2 \\ \downarrow \end{matrix}$	$A_1 B_1 + C_1$	$A_0 B_0$
	(C_2)	(C_1)
	Y_3	Y_2
	Y_1	Y_0

$$B_1 \times 2^1 \times A_0 \times 2^0 \quad \text{I.}$$

$$(B_1 A_0) \times 2^1$$

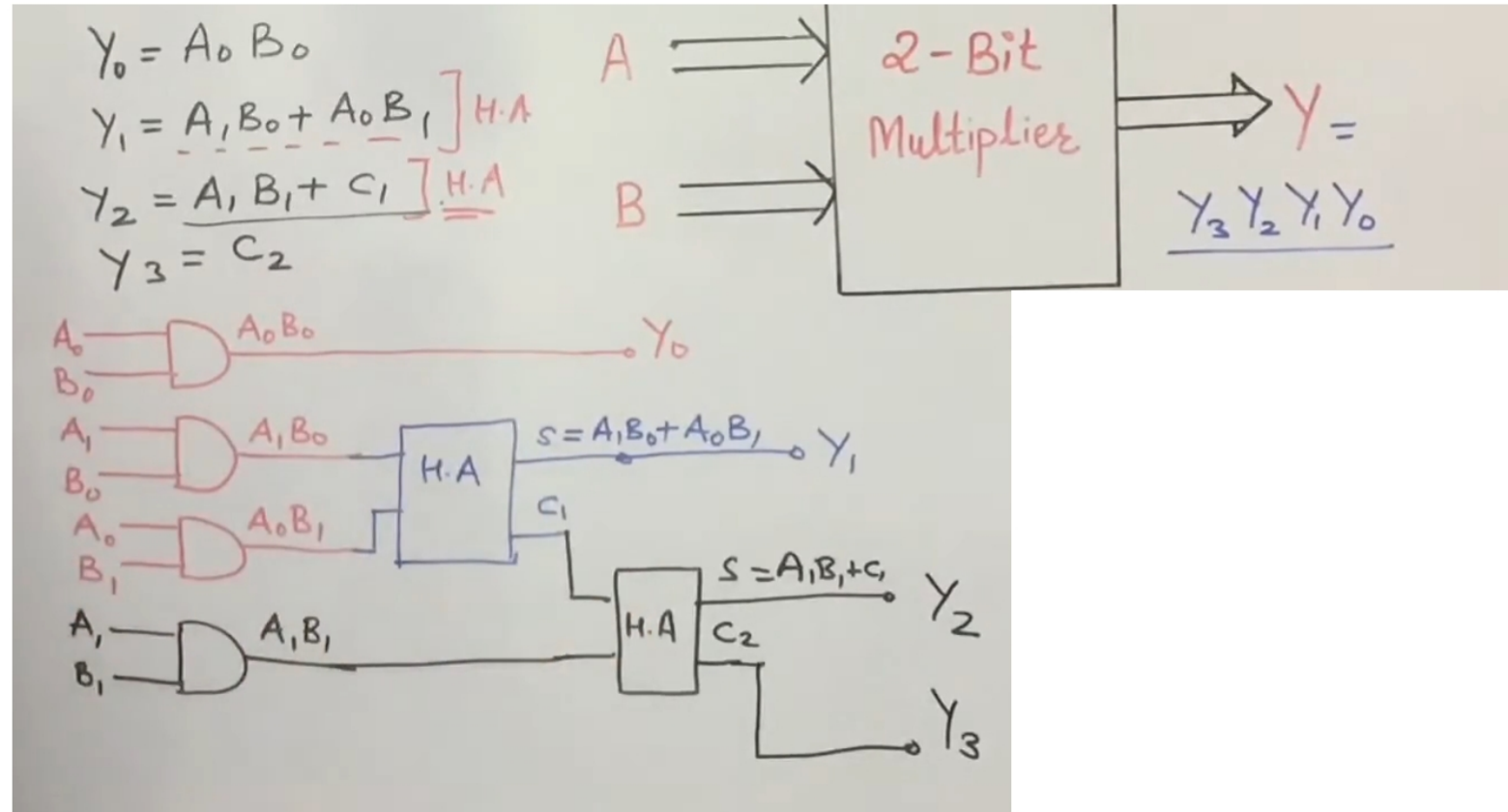
$$B_1 \times 2^{\textcircled{1}} \times A_1 \times 2^{\textcircled{1}}$$

$$= (B_1 A_1) \times 2^2$$



2-BIT MULTIPLIER

Circuit diagram with Half Adders



2-BIT MULTIPLIER

$$A = \overset{2^2}{A_2} \overset{2^1}{A_1} \overset{2^0}{A_0}$$

$$A_0 B_0$$

$$A_0 B_0$$

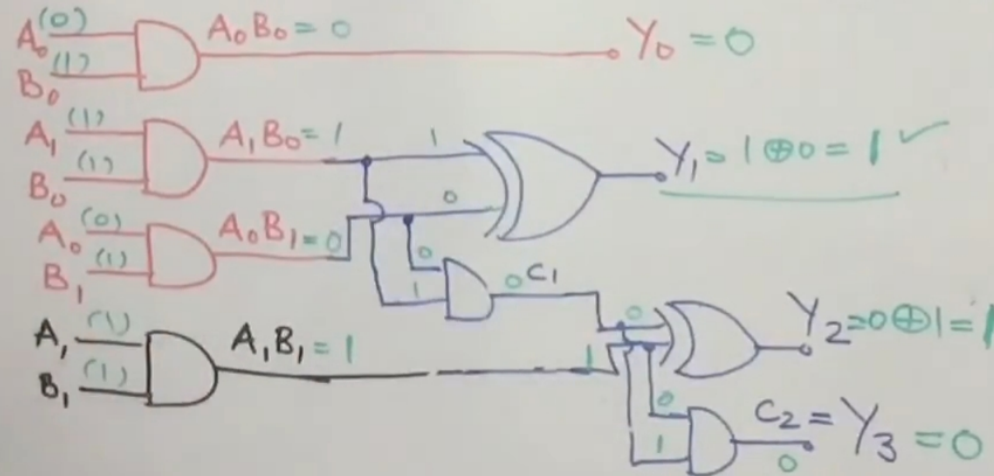
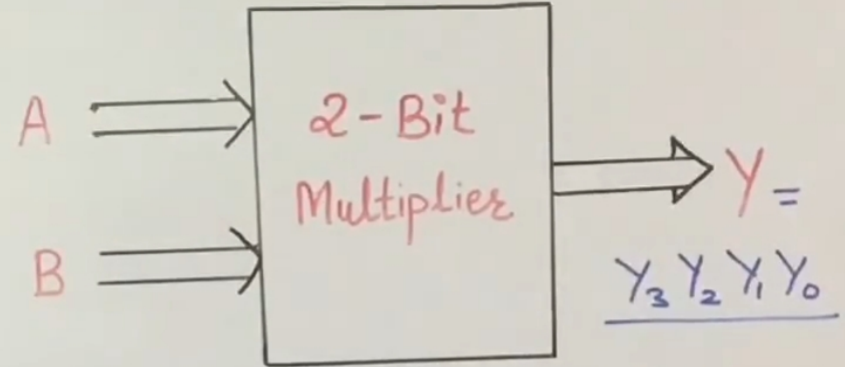
$$Y_0$$

$$Y_0 = A_0 B_0$$

$$Y_1 = A_1 B_0 + A_0 B_1 \quad \text{H.A.}$$

$$Y_2 = A_1 B_1 + C_1 \quad \text{H.A.}$$

$$Y_3 = C_2$$



$$\begin{array}{r} A = 10 \quad (2) \\ B = 11 \quad (3) \\ \hline 10 \\ 10x \\ \hline 0110 \quad (6) \\ \hline Y_3 Y_2 Y_1 Y_0 \end{array}$$